

Prescribed Isolated Intervals in Higher Fold Sumsets

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Abstract

We give an explicit construction of a finite set of integers whose h -fold sumset contains any prescribed finite target set and whose remaining elements are isolated. As a consequence, every sufficiently separated family of intervals can be prescribed as exactly the nontrivial intervals in an h -fold sumset. This answers the multi-interval questions numbered Problems 14–16 in Nathanson’s June 8, 2026 revision of *Problems in additive number theory, VII*. In fact the required gap threshold is the minimal value $N_q = n + 2$ allowed by the formulation, and the construction has $|A| = 2q(n + 1)$.

The first proof uses a separated-label principle that reduces the problem to short integer relations among labels. We then replace the initial exponential labels by finite moment labels. For fixed h , this gives a diameter bound polynomial in the number of prescribed target points and independent of their absolute location after centering. For double sumsets, a shifted-Sidon label instance improves the diameter bound to quadratic, and a simple Sidon difference-counting argument shows that this quadratic order is unavoidable within the Sidon-label method. The main separation, interval-family, cardinality, translation, diameter, and shifted-Sidon arguments have been checked in Lean 4 with Mathlib.

1 Introduction

Nathanson’s problem list on additive h -bases for intervals asks how much control is possible over the intervals contained in finite sumsets. In a June 8, 2026 revision, the single-interval double-sumset case is constructed directly, and the remaining multi-interval questions ask whether one can prescribe a separated set

$$C = \{c_1 < \cdots < c_q\}$$

of starting points so that the only intervals of length n in $2A$ start at the elements of C , with a stronger version forbidding all additional proper intervals. The same revision asks for the corresponding higher-fold problem when $h \geq 3$.

The point of this note is that the multi-interval problem is more naturally a finite-target problem. Let

$$S = \bigcup_{i=1}^q [c_i, c_i + n].$$

It is enough to construct A so that

$$hA = S \dot{\cup} E,$$

where every point of E is separated from S and from every other point of E by at least two. Then the maximal nontrivial intervals in hA are exactly the maximal nontrivial intervals already present in S . The construction below proves this for every $h \geq 2$ and, more generally, for an arbitrary finite target set $S \subseteq \mathbb{Z}$.

The mechanism is a tagged cancellation construction. To each target point t_r we attach a large label a_r and put two generators into A ,

$$t_r + (h-1)a_r \quad \text{and} \quad -a_r.$$

The intended representation of t_r is the matched sum consisting of one positive generator and $h-1$ copies of the corresponding negative generator. All other h -term sums carry a nonzero label. If the label family has no short relations, then these exceptional sums are far apart.

The first version uses powers and has exponential diameter. The later sections replace the powers by finite moment labels, giving polynomial diameter for fixed h , and by shifted-Sidon labels when $h=2$, giving quadratic diameter. The Sidon section also includes a lower bound showing that quadratic diameter is the best possible order for this Sidon-label strategy. These quantitative refinements are not needed merely to answer the existence questions, but they make the construction less ad hoc and identify where the real optimization problem begins.

2 A short separation lemma

For an integer vector $v = (v_0, \dots, v_{s-1})$, write $\|v\|_1 = \sum_r |v_r|$.

Lemma 1 (Positional separation). *Let $B, M \geq 1$ and $a_r = MB^r$ for $0 \leq r < s$. If $\lambda, \mu \in \mathbb{Z}^s$, $\lambda \neq \mu$, and $\|\lambda\|_1 + \|\mu\|_1 \leq B$, then*

$$\left| \sum_{r=0}^{s-1} (\lambda_r - \mu_r) a_r \right| \geq M.$$

Proof. Put $z = \lambda - \mu$ and let R be the largest index with $z_R \neq 0$. We have $\|z\|_1 \leq B$. If $R=0$, the result is immediate. If $R \geq 1$, then

$$\begin{aligned} \left| \sum_{r=0}^{s-1} z_r a_r \right| &\geq M \left(|z_R| B^R - \sum_{r < R} |z_r| B^r \right) \\ &\geq M (B^R - (B-1)B^{R-1}) \geq M. \end{aligned}$$

Here the sum of the absolute values of the lower coefficients is at most $B - |z_R| \leq B-1$. $\square$

Proposition 1 (Separated-label principle). *Let $h \geq 2$, $d \geq 1$, and let $S = \{t_0, \dots, t_{m-1}\} \subseteq \mathbb{Z}$ be nonempty. Put*

$$H = h(h-1), \quad T = \max_{t \in S} |t|.$$

Suppose that $M \geq 2hT + d$ and that integers $a_0, \dots, a_{m-1}$ satisfy

$$\left| \sum_{r=0}^{m-1} (\lambda_r - \mu_r) a_r \right| \geq M \tag{1}$$

whenever $\lambda, \mu \in \mathbb{Z}^m$ are distinct and

$$\|\lambda\|_1 + \|\mu\|_1 \leq 2H.$$

Then

$$A = \{t_r + (h-1)a_r : 0 \leq r < m\} \cup \{-a_r : 0 \leq r < m\}$$

has $|A| = 2m$ and

$$hA = S \dot{\cup} E,$$

where distinct elements of E are at distance at least d from one another and every element of E is at distance at least d from S .

Proof. Attach to the generator $t_r + (h-1)a_r$ the label vector $(h-1)e_r$ and offset t_r . Attach to the generator $-a_r$ the label vector $-e_r$ and offset 0. The integer value of a tagged generator is its offset plus the dot product of its label vector with $(a_0, \dots, a_{m-1})$.

We first check that the displayed union defining A is disjoint and has the claimed cardinality. Distinct tagged generators have distinct label vectors. The sum of the ℓ^1 norms of two generator labels is at most $2(h-1) \leq 2H$, and their offsets differ by at most $2T$. Since $M \geq 2hT + d > 2T$, the separation hypothesis (1) implies that two distinct tagged generators cannot have the same integer value. Hence $|A| = 2m$.

Now consider an arbitrary h -term sum from A . Let p_r be the number of times the positive generator $t_r + (h-1)a_r$ is used, and let u_r be the number of times the negative generator $-a_r$ is used. Thus

$$\sum_r (p_r + u_r) = h.$$

The label coefficient of this representation is

$$v_r = (h-1)p_r - u_r.$$

Its norm is bounded by

$$\sum_r |v_r| \leq (h-1) \sum_r p_r + \sum_r u_r \leq h(h-1) = H,$$

and the offset $\sum_r p_r t_r$ has absolute value at most hT .

The target points are exactly the sums with zero label. Indeed, if $v = 0$, then $(h-1)p_r = u_r$ for every r . Therefore

$$\sum_r (p_r + u_r) = h \sum_r p_r = h,$$

so $\sum_r p_r = 1$. There is exactly one index r with $p_r = 1$, and then $u_r = h-1$ while all other counts vanish. The corresponding sum is

$$(t_r + (h-1)a_r) + (h-1)(-a_r) = t_r.$$

Conversely, this matched representation gives every t_r . Thus the zero-label part of hA is precisely S .

It remains to see that exceptional sums cannot cluster. A nonzero label vector determines its multiset of terms. Suppose two h -term multisets with counts (p_r, u_r) and (p'_r, u'_r) have the same label vector. Then

$$u_r - u'_r = (h-1)(p_r - p'_r).$$

Summing this identity over r and using the fact that both multisets have h terms gives $\sum_r p_r = \sum_r p'_r$. If the positive count vectors were not equal, choose an index with $p_r > p'_r$. Then $p_r \geq 1$ and the displayed identity gives $u_r \geq h-1$. These already use all h terms, so the first multiset is a matched zero-label representation, contradicting the assumption that the common label vector is nonzero. Hence $p = p'$ and then also $u = u'$.

Consequently, two distinct exceptional sums have distinct nonzero label vectors. Also, every exceptional sum has a label vector distinct from the zero label of a target point. For two exceptional sums, or for an exceptional sum and a target point, the two label vectors have combined norm at most $2H$. Equation (1) gives label separation at least M , while the two offsets together can change the distance by at most $2hT$. The remaining distance is at least

$$M - 2hT \geq d.$$

This proves the required decomposition and isolation. □

3 The construction

Theorem 1 (Finite target realization). *Let $h \geq 2$, $d \geq 1$, and let $S \subseteq \mathbb{Z}$ be a nonempty finite set with $m = |S|$. There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2m$ such that*

$$hA = S \dot{\cup} E,$$

where distinct elements of E are at distance at least d from one another and every element of E is at distance at least d from S .

Proof. Enumerate $S = \{t_0, \dots, t_{m-1}\}$ and put

$$T = \max_{t \in S} |t|, \quad M = 2hT + d, \quad B = 2h(h-1), \quad a_r = MB^r.$$

Lemma 1 verifies (1) for these labels: if the two coefficient vectors have combined ℓ^1 norm at most $B = 2h(h-1)$, then their positional values differ by at least M . Proposition 1 therefore gives the required set

$$A = \{t_r + (h-1)a_r : 0 \leq r < m\} \cup \{-a_r : 0 \leq r < m\}.$$

□

Theorem 2 (Prescribed isolated intervals). *Let $h \geq 2$, $n \geq 1$, $d \geq 2$, $q \geq 1$, and*

$$C = \{c_1 < \dots < c_q\} \subseteq \mathbb{Z}$$

with $c_{i+1} - c_i \geq n + 2$ for all i . There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2q(n+1)$ such that

$$hA = S \dot{\cup} E, \quad S = \bigcup_{i=1}^q [c_i, c_i + n],$$

where

$$|x - y| \geq d$$

whenever $x, y \in E$ are distinct or $x \in E$ and $y \in S$. Consequently, the maximal intervals in hA of length at least one are exactly the intervals $[c_i, c_i + n]$.

Proof. Because $c_{i+1} - c_i \geq n + 2$, the intervals $[c_i, c_i + n]$ are disjoint and separated from one another by at least one missing integer. Hence $|S| = q(n+1)$. Apply Theorem 1 to this finite target set S . It gives $|A| = 2q(n+1)$, the decomposition $hA = S \dot{\cup} E$, and the required distance bounds. Since $d \geq 2$ and the intervals in S are separated by at least one missing integer, no exceptional point can extend a target interval or form a new nontrivial interval. □

Corollary 1 (Single interval specialization). *For all $h \geq 2$, $c \in \mathbb{Z}$, $n \geq 1$, and $d \geq 2$, there is a set $A \subseteq \mathbb{Z}$ with $|A| = 2n + 2$ such that $[c, c + n] \subseteq hA$, every element of $hA \setminus [c, c + n]$ is at distance at least d from every other such element and from $[c, c + n]$, and the longest interval in hA has length n .*

Proof. Apply Theorem 2 with $C = \{c\}$. □

Corollary 2 (Nathanson's revised multi-interval problems). *In the notation of Problems 14–16 in Nathanson's June 8, 2026 revision, Problems 14 and 15 are true, and Problem 16 is true for every $h \geq 3$. More precisely, one can take the threshold*

$$N_q = n + 2$$

and

$$k_C = 2q(n+1).$$

Proof. The hypotheses in the revised problems include $2 \leq n \leq N_q - 2$, so $N_q = n + 2$ is the smallest possible threshold compatible with that formulation. If $c_{i+1} - c_i \geq n + 2$, then Theorem 2 with $d = 2$ gives

$$hA = S \dot{\cup} E, \quad S = \bigcup_{i=1}^q [c_i, c_i + n],$$

where every point of E is isolated from every other point of hA .

For $h = 2$, the only intervals of length n in $2A$ are the displayed intervals $[c_i, c_i + n]$, proving Problem 14. Since every element outside S is isolated and at distance at least two from S , every interval of length one in $2A$ lies inside one of the intervals $[c_i, c_i + n]$, proving the stronger Problem 15. The same argument with the prescribed value $h \geq 3$ proves Problem 16. $\square$

Remark 1 (Diameter bound). *For*

$$S = \bigcup_{i=1}^q [c_i, c_i + n], \quad T = \max_{s \in S} |s|,$$

the construction in Theorem 2 satisfies

$$\text{diam}(A) \leq h(2hT + d)(2h(h-1))^{q(n+1)-1} + T.$$

The bound is intentionally simple rather than optimized.

4 A polynomial diameter refinement

The power labels in Theorem 2 make the first construction transparent, but their size is exponential in the number of target points. The following finite moment encoding gives a polynomial bound for fixed h .

Lemma 2 (Moment separation). *Let $m, L \geq 1$, put*

$$R = \max(1, m-1), \quad K = LR^{L-1}, \quad Q = LK = L^2R^{L-1},$$

and define

$$b_r = \sum_{j=0}^{L-1} r^j Q^j \quad (0 \leq r < m).$$

If $z = (z_0, \dots, z_{m-1}) \in \mathbb{Z}^m$ is nonzero and $\|z\|_1 \leq L$, then

$$\left| \sum_{r=0}^{m-1} z_r b_r \right| \geq 1.$$

Proof. For $0 \leq j < L$, put

$$\mu_j = \sum_{r=0}^{m-1} z_r r^j.$$

Not all μ_j vanish. Indeed, if the support of z has size s , then $s \leq L$, and the first s equations $\mu_j = 0$ would form an invertible Vandermonde system on the distinct support indices.

Since $|\mu_j| \leq LR^{L-1} = K$, the moment vector has ℓ^1 norm at most $LK = Q$. A nonzero integer coefficient vector with ℓ^1 norm at most Q cannot evaluate to zero in base Q : if J is its largest nonzero index, the leading term has absolute value at least Q^J , while all lower terms together have absolute value at most $(Q-1)Q^{J-1}$ when $J \geq 1$; the case $J = 0$ is immediate. Applying this positional domination to $(\mu_0, \dots, \mu_{L-1})$ proves the claim. $\square$

Theorem 3 (Polynomial diameter realization). *Let $h \geq 2$, $d \geq 2$, and let $S \subseteq \mathbb{Z}$ be a nonempty finite set with $m = |S|$. There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2m$ such that*

$$hA = S \dot{\cup} E,$$

where

$$|x - y| \geq d$$

whenever $x, y \in E$ are distinct or $x \in E$ and $y \in S$.

More precisely, let $\tau \in h\mathbb{Z}$ minimize

$$T = \max_{t \in S} |t - \tau|.$$

Put

$$L = 2h(h-1), \quad R = \max(1, m-1), \quad Q = L^2 R^{L-1},$$

and

$$P_h(m) = \sum_{j=0}^{L-1} (m-1)^j Q^j.$$

Then A can be chosen with

$$\text{diam}(A) \leq h(2hT + d)P_h(m) + T.$$

For fixed h , the factor $P_h(m)$ is bounded by a polynomial in m of degree $L(L-1)$.

Proof. Enumerate $S = \{t_0, \dots, t_{m-1}\}$ and use the parameters in Lemma 2. Set

$$M = 2hT + d, \quad a_r = M \sum_{j=0}^{L-1} r^j Q^j,$$

and first construct the centered set

$$A_0 = \{t_r - \tau + (h-1)a_r : 0 \leq r < m\} \cup \{-a_r : 0 \leq r < m\}.$$

Finally, put

$$A = A_0 + \tau/h.$$

If $\lambda \neq \mu$ and $\|\lambda\|_1 + \|\mu\|_1 \leq L$, then $z = \lambda - \mu$ is nonzero and $\|z\|_1 \leq L$. Lemma 2, after multiplication by M , therefore verifies (1) for the centered target set. Proposition 1 gives $|A_0| = 2m$, the matching-label decomposition, and the required distance bounds. Translating A_0 by τ/h translates hA_0 by τ and does not change its diameter.

The labels a_r are strictly increasing. Every element of A_0 lies between $-a_{m-1}$ and $T + (h-1)a_{m-1}$. Consequently,

$$\text{diam}(A) = \text{diam}(A_0) \leq ha_{m-1} + T = h(2hT + d)P_h(m) + T.$$

For $m \geq 2$, the displayed expression for $P_h(m)$ is a polynomial of degree $L(L-1)$; the case $m = 1$ is immediate. $\square$

Corollary 3 (Translation-invariant interval bound). *Under the hypotheses of Theorem 2, let*

$$m = q(n+1), \quad D = c_q + n - c_1.$$

There is a set A with the same isolated interval pattern and $|A| = 2q(n+1)$ whose diameter is polynomial in m for fixed h , with

$$T \leq \left\lceil \frac{D+h}{2} \right\rceil$$

in the bound of Theorem 3. In particular, the bound is independent of the absolute location of the prescribed intervals.

Proof. Apply Theorem 3 to $S = \bigcup_i [c_i, c_i + n]$. A multiple τ of h can be chosen within $h/2$ of the midpoint of $[c_1, c_q + n]$, giving the stated bound for T . $\square$

Theorem 4 (Quadratic diameter for double sumsets). *Let $d \geq 2$, and let $S \subseteq \mathbb{Z}$ be a nonempty finite set with $m = |S|$. There is an explicitly constructible set $A \subseteq \mathbb{Z}$ with $|A| = 2m$ such that*

$$2A = S \dot{\cup} E,$$

where distinct elements of E are at distance at least d from one another and every element of E is at distance at least d from S .

More precisely, let $\tau \in 2\mathbb{Z}$ minimize

$$T = \max_{t \in S} |t - \tau|,$$

and let $p \geq m$ be an odd prime. For $0 \leq r < m$, let ρ_r be the least nonnegative residue of r^2 modulo p , and put

$$b_r = 2p^2 + 2pr + \rho_r.$$

Then A can be chosen with

$$\text{diam}(A) \leq 8(4T + d)p^2 + T.$$

For $m \geq 2$, Bertrand's postulate allows $p < 2m$, giving a quadratic bound in m . Since p is an integer, the resulting explicit estimate is

$$\text{diam}(A) \leq 8(4T + d)(2m - 1)^2 + T.$$

Proof. First observe that $\{b_0, \dots, b_{m-1}\}$ is a Sidon set, with repetitions allowed in pair sums. Indeed, suppose

$$b_i + b_j = b_k + b_\ell.$$

Reducing modulo $2p$ gives $\rho_i + \rho_j = \rho_k + \rho_\ell$, because both sides lie between 0 and $2p - 2$. It follows that $i + j = k + \ell$. Reducing the residue identity modulo p also gives

$$i^2 + j^2 \equiv k^2 + \ell^2 \pmod{p}.$$

Since p is odd, we obtain $ij \equiv k\ell \pmod{p}$. Thus the two pairs are the roots of the same quadratic polynomial over $\mathbb{F}_p$. The unordered pairs $\{i, j\}$ and $\{k, \ell\}$ agree modulo p , hence agree as integer pairs because all indices lie in $[0, p)$.

Moreover,

$$2p^2 \leq b_r < 4p^2.$$

Consequently, an equality between two sums of the b_r involving at most four terms in total must have the same number of terms on both sides: one term is strictly smaller than any sum of two terms. Equal one-term sums are trivial, and equal two-term sums are trivial by the Sidon property. Therefore

$$\sum_r z_r b_r \neq 0$$

for every nonzero integer vector z with $\|z\|_1 \leq 4$.

Set $M = 4T + d$, enumerate $S = \{t_0, \dots, t_{m-1}\}$, and apply Proposition 1 to the centered targets $t_r - \tau$ and the labels $a_r = Mb_r$. Finally translate the resulting basis by $\tau/2$. Since $b_r < 4p^2$, its diameter is at most

$$2M \max_r b_r + T \leq 8(4T + d)p^2 + T.$$

The same construction has quadratic diameter from below. Indeed, translation does not change diameter, the negative generators contain $-Mb_0$ and $-Mb_{m-1}$, and

$$b_{m-1} - b_0 = 2p(m - 1) + \rho_{m-1} \geq 2m(m - 1).$$

Consequently,

$$\text{diam}(A) \geq 2Mm(m-1).$$

□

Proposition 2 (A Sidon-label diameter barrier). *Let $B = \{b_1, \dots, b_m\} \subseteq \mathbb{Z}$ be a Sidon set, where repetitions are allowed in pair sums. Then*

$$\text{diam}(B) \geq \binom{m}{2}.$$

Consequently, any tagged construction that contains the generators $-Mb_i$ has diameter at least $M\binom{m}{2}$.

Proof. Order the labels so that $b_1 < \dots < b_m$. The $\binom{m}{2}$ positive differences $b_j - b_i$, with $i < j$, are distinct. Indeed, an equality $b_j - b_i = b_\ell - b_k$ gives $b_j + b_k = b_\ell + b_i$, and the Sidon property forces the two ordered index pairs to agree. Every positive difference is at most $\text{diam}(B)$, proving the first claim. The second follows because the generators $-Mb_i$ already have diameter $M\text{diam}(B)$. □

Remark 2 (Elementary lower bounds). *Let $A \subseteq \mathbb{Z}$ be finite and nonempty, put $k = |A|$, and suppose that $S \subseteq hA$ has m elements. Then*

$$m \leq \binom{k+h-1}{h}, \quad \text{diam}(A) \geq \frac{\text{diam}(S)}{h}.$$

Indeed, the first inequality counts unordered h -term multisets from A , and the second follows from

$$\text{diam}(hA) = h \text{diam}(A).$$

Thus Theorem 3 supplies a polynomial diameter bound for fixed h , while Theorem 4 gives a quadratic bound when $h = 2$. Proposition 2 shows that the quadratic order is unavoidable within the Sidon-label method. Optimizing either the cardinality or the degree for $h \geq 3$ remains open.

5 Relation with the single-interval construction

Nathanson’s June 8 revision contains a direct construction for one prescribed interval in a double sumset. The special case $q = 1$ and $h = 2$ of Corollary 1 gives another route to the same type of single-interval conclusion, but this is not the main point of the present paper. The label construction is designed to make independent cancellations for many target points coexist without producing additional adjacent exceptional sums. This is what allows the same argument to treat the multi-interval questions and the higher-fold case uniformly.

The quantitative refinements also belong to the finite-target viewpoint. Moment labels make the construction polynomial in the number of target points for fixed h , after centering the target set. For $h = 2$, shifted Sidon labels give a quadratic diameter bound, and Proposition 2 shows that this order is forced by the Sidon-label mechanism itself.

6 Reproducibility

The accompanying Python package constructs A , computes hA , checks all maximal consecutive runs, records the minimum exceptional spacing, and emits JSON certificates. It also includes exact small-parameter computations cross-checked against [OEIS A001208](#) and [OEIS A001209](#). The arithmetic separation, bounded-offset, matched-count, tagged-representation, exceptional-count uniqueness, generic set-level isolation, interval-enumeration, maximal-run, and exact-cardinality arguments have also been checked in Lean 4 with Mathlib. The resulting Lean theorem automatically selects the target radius and

substitutes the displayed values of M and B . The finite moment-label Vandermonde argument, translation rule, closed diameter estimate, exact cardinality, midpoint-centering lemma, and interval-family theorem have also been formalized. For the shifted-Sidon refinement, Lean checks the size bounds and the reduction from order-four short-relation freedom to set-level isolation, including the explicit pair-sum argument over $\mathbb{F}_p$. It also checks translation invariance, quadratic diameter upper and lower bounds, tightened Bertrand-prime selection, exact cardinality after translation, and the final separated-interval wrapper. The midpoint translation, radius, and scale are selected automatically from an enclosing interval.

References

- [1] Melvyn B. Nathanson, *Problems in additive number theory, VII: The structure of additive h -bases for n* , preprint, 2026. [arXiv:2605.26425](https://arxiv.org/abs/2605.26425).